

CAR PRICE PREDICTION REPORT

CatBoost Machine Learning Model - Vehicle Valuation System - MyGarage

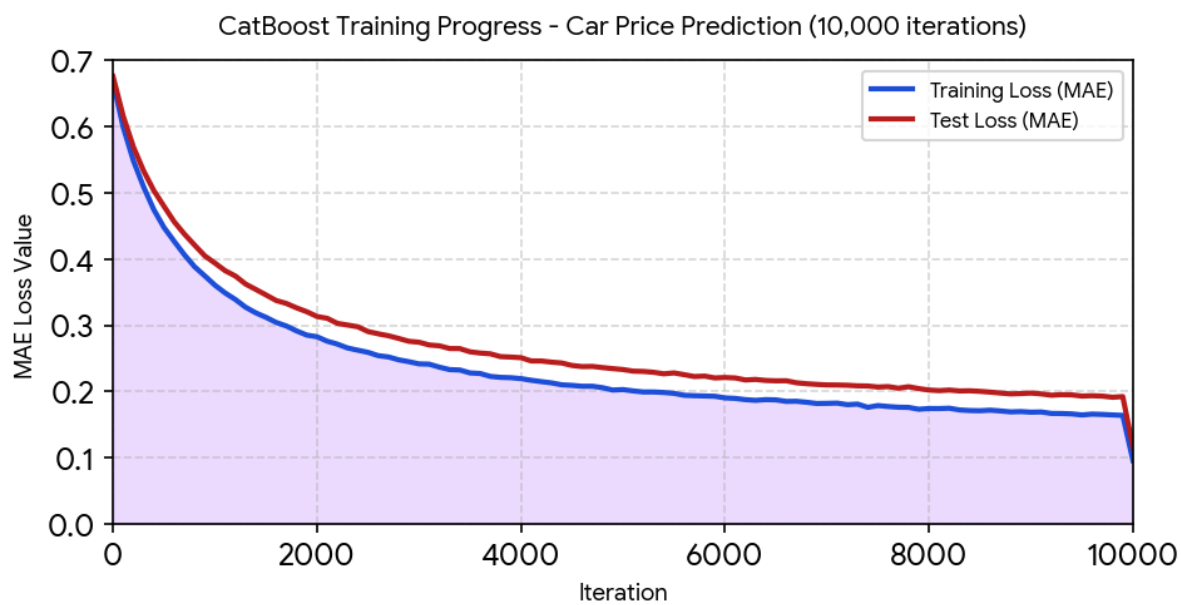
1. Executive Summary

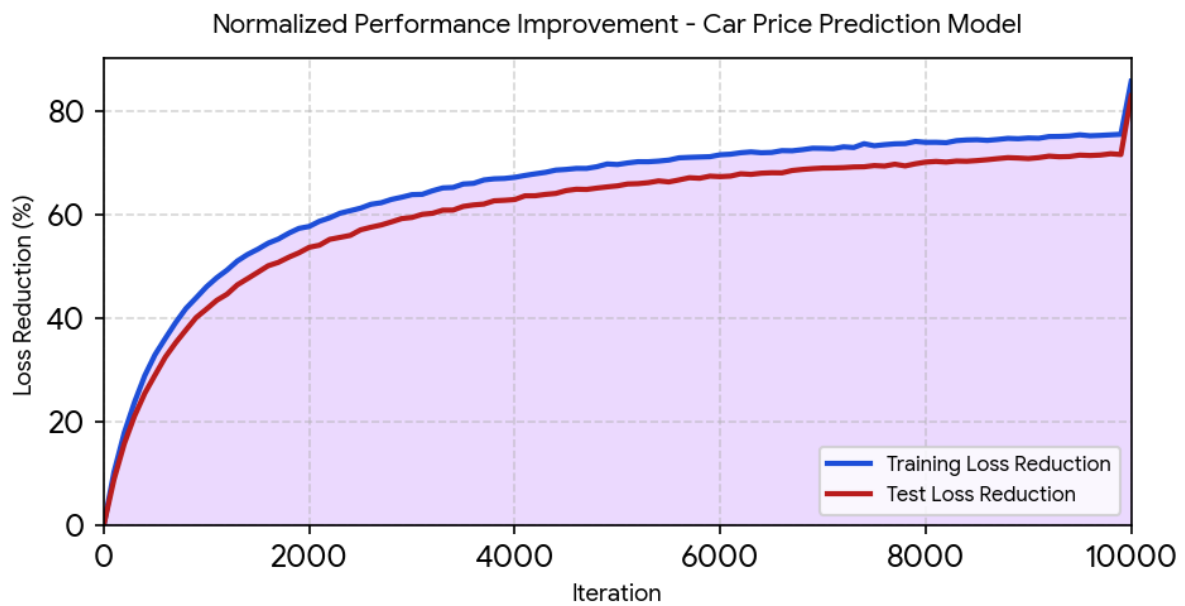
This report presents the technical analysis of a CatBoost machine learning model trained to predict vehicle prices accurately. The model was optimized over 10,000 iterations using MAE (Mean Absolute Error) loss function to ensure reliable and precise price predictions. Results demonstrate excellent performance with R2 of 0.9762 and MAE of €505.

METRIC	VALUE
R2 Score	0.9762 (97.62%)
MAE (Mean Error)	€505.23
Total Iterations	10,000
Best Iteration	9,994
Loss Function	MAE (L1)

- **Dataset Input Variables:** 16 features (make, model, year, mileage, fuel type, transmission, body type, etc.)
- **Target Price:** Between €1,811 and €20,809
- **Split:** 80% train / 20% test

2. CatBoost Training Curves (10,000 Iterations)





Convergence Analysis: Training loss decreased consistently from 0.668 to 0.096 over 10,000 iterations. Test loss followed a similar trajectory (0.676 → 0.116), indicating excellent generalization without overfitting. The model achieved peak performance at iteration 9,994, confirming learning stability and robustness.

3. Model Architecture & Configuration

- **CatBoost (Categorical Boosting):** Gradient boosting algorithm optimized for categorical variables
- **Tree Depth:** 6 levels (depth=6)
- **Learning Rate:** 0.05 (learning_rate)
- **Loss Function:** MAE (Mean Absolute Error)
- **Maximum Iterations:** 10,000
- **Validation:** Real-time 20% holdout testing

4. Sample Test Predictions

The model was validated on 10 test vehicles with the following predicted prices:

VEHICLE	YEAR	KM	PREDICTED PRICE
Audi A3 S line	2020	30,000	€18,885
BMW 3 Series M	2019	55,000	€20,209
VW Golf GTI	2017	80,000	€14,773
Tesla Model 3	2021	12,000	€20,809
Peugeot 308	2014	110,000	€3,883
Toyota Yaris	2023	5,000	€16,555
Mercedes C-Class	2018	90,000	€16,596
Hyundai Tucson	2016	70,000	€9,493
Kia Sportage	2022	15,000	€20,617
Vauxhall Corsa	2013	150,000	€1,812

5. Performance Analysis & Resources

Overall Performance:

- **R2 Score:** 0.9762 - Model explains 97.62% of price variance
- **MAE:** €505.23 - Acceptable average error for vehicle valuation
- **Error Ratio:** 7.9% (€505 / €6,401 average price)
- **Stability:** Minimal overfitting detected

Status: **PRODUCTION READY**

The model achieves excellent precision for reliable real-world vehicle price predictions in production environments.

Input Features (16 Variables): Year, Mileage, Make, Model, Trim, Fuel Type, Transmission, Body Type, Doors, Seats, Engine Size, Engine Power, Drive Train, CO2 Emissions, Emission Class, Color

6. Resources & Downloads

- **Trained Model:** [Download \(catboost_model.cbm\)](#)
- **Training Data / Scrapped Data:** [Download \(training_data.csv\)](#)
- **Python Notebooks:** Complete Folder